# Adam G G Smith

**Postdoctoral Research Associate in Numerical Modelling**
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## Personal Statement

I develop new quantitative methods to leverage large topographic data sets, combined with field data and lab work, to resolve fundamental questions related to Earth surface processes.

## Experience/Education

**Postdoctoral Research Associate in Numerical Modelling, University of Glasgow (2024 – present)**
Project – Landscape evolution modelling to understand bio-geosphere interactions in future climate scenarios
Advisor – Prof. Todd Ehlers

**PhD in Tectonic Geomorphology**, **UCL/Birkbeck, University of London (2020 – 2024)**
Title - 'Exploring Earth's Dynamic Surface with Topographic Inverse Methods'
Supervisors - Prof. Andrew Carter and Dr Matthew Fox

**MSci in Geology, Imperial College London (2016 – 2020)**
First class honours

## Awards

**Bernie Smith Award** (Sep 2024) – For the best postgraduate talk at BSG Annual Meeting 2024
**E. Ambrosiodou Prize** (Jun 2019 - £1000) – For best 3rd year mapping project (dissertation)
**Earth Science and Engineering Medal** (Jun 2020) – For outstanding student performance
**3rd Year Dean's List** (Jun 2019) – For the top 10% of the year at Imperial College London

## Grants

**National Environmental Isotope Facility Grant** (Dec 2025 – 56 Samples) – Catchment-wide denudation rates in Scotland from in situ-produced 10Be and 26Al. Co-PI (30%) with Mirjam Schaller, Todd Ehlers and Derek Fabel.

**Royal Society International Exchange Grant** (Nov 2025 - £12000) – Modelling the Mechanisms Underlying Permafrost Landscape-Vegetation Interactions and Their Impact on Riverine Sediment and Carbon Transport Dynamics. Co-Investigator.

**BSG Postgraduate Research Grant** (Mar 2024 - £1580) – Funding for fieldwork on the Isle of Jura, investigating the dynamics of river incision during post-glacial rebound

**UCL-Yale doctoral exchange** (Jan 2023 - £6000) – Funding for 3 month exchange at Yale University with Prof. Mark Brandon

## Publications

**Smith, A.G.G.**, Fox, M., Akinwale, O., Rugen, E.J. 2026. How does lithology influence fluvial erodibility? A study of knickpoint retreat in small bedrock rivers on the Isle of Jura. *Earth Surface Processes and Landforms.* doi.org/10.1002/esp.70387

Fox, M., **Smith, A.G.G.**, Vermeesch, P., Gallagher, K., Carter, A. 2026. Uncertainty in helium diffusion in zircon limits thermochronologic resolution: application to the Great Unconformity. *Geochronology*. doi.org/10.5194/gchron-8-463-2026

Han, X., **Smith, A.G.G.**, Dai, J., Fox, M. 2026. Reply to: Himalaya rivers reflect rainfall patterns, not river piracy. *Nature Geoscience*. doi.org/10.1038/s41561-026-02061-7

***Smith, A.G.G.**, Fox, M., Miller, S.R., Morriss, M.C., Anderson, L. 2026. A lithospheric drip evidenced by relict topography of the Uinta Mountains could have triggered Green and Colorado River integration. *Journal of Geophysical Research: Earth Surface.* doi.org/10.1029/2025JF008733. ***Most read article in JGR: Earth Surface of all time.** Top 5% of all research outputs on altmetric, covered by 59 news outlets, including CNN, Le Monde and Al Jazeera.

Lanari, R., Bonini, M., Sembroni, A., Papeschi, S., Del Ventisette, C., **Smith, A.G.G.**, Lupi, M., Montanari, D. 2025. Topographic signature of magmatic emplacement at depth: The case of the Larderello-Travale Geothermal area (Northern Apennines, Italy). *Earth and Planetary Science Letters.* doi.org/10.1016/j.epsl.2025.119290

**Smith, A.G.G.** and Fox, M. 2024. When does the concavity index constrain stream power parameters? *Journal of Geophysical Research: Earth Surface*, 129 (9), e2023JF007584. doi.org/10.1029/2023JF007584

*Han, X., Dai, J., **Smith, A.G.G.**, Xu, S., Liu, B., Wang, C., Fox, M. 2024. Recent uplift of Chomolungma enhanced by river drainage piracy. *Nature Geoscience*. doi.org/10.1038/s41561-024-01535-w. ***Top 5% of all research outputs on altmetric**, covered by 416 global news outlets including the BBC, CNN and NY Times

Fox, M., Clinger, A., **Smith, A.G.G.**, Cuffey, K., Shuster, D., Herman, F., 2024. Antarctic Peninsula glaciation patterns set by landscape evolution and dynamic topography. *Nature Geoscience.* doi.org/10.1038/s41561-023-01336-7.

Rugen, E.J., Pastore, G., Vermeesch, P., Spencer, A., Webster, D., **Smith, A.G.G.**, Carter, A., Shields, G. 2024. Glacially influenced provenance and Sturtian affinity revealed by detrital zircon U-Pb ages from the Port Askaig Formation, Dalradian Supergroup. *Journal of the Geological Society*. doi.org/10.1144/jgs2024-029.

**Smith, A.G.G.**, Fox, M., Moore, J.R., Miller, S.R., Goren, L., Morriss, M.C., Carter, A., 2024. One million years of climate driven rock uplift on the Wasatch Fault revealed by fluvial topography. *American Journal of Science, 324.* 1. doi.org/10.2475/001c.92194.

Fu, H., Kereszturi, G., Cheng, Q., Wang, R., **Smith, A.G.G.**, 2024. Deciphering Differential Exhumation in the Gangdese Orogen Using Exposed Porphyry Alteration Systems and Geomorphic Analysis. *Geological Society of America Bulletin.* doi.org/10.1130/B37086.1

Fox, M., Hoseason, T., Bernard, T., Sinclair, H**., Smith, A.G.G.**, 2023. Bedload-Bedrock Contrasts Form Enigmatic Low-Relief Surfaces of the Pyrenees. *Geophysical Research Letters*. 50 (6). doi.org/10.1029/2022GL101995

**Smith, A.G.G.**, Fox, M., Schwanghart, W., Carter, A., 2022. Comparing methods for calculating channel steepness index. *Earth-Science Reviews*, *227.* 103970. doi.org/10.1016/j.earscirev.2022.103970

**Smith, A.G.G.**, Ehlers, T.A., Hurst, M.D, Lai, J. *In Review at Science*. Climate Controls Bedrock River Concavity.

**Smith, A.G.G.**, Hurst, M.D, Ehlers, T.A. *In Review at Earth and Planetary Science Letters*. Discerning Tectonic vs Climatic Controls of Topography.

Lanari, R., **Smith, A.G.G.**, Stumpp, D., Bonini, M., Del Ventisette, C., Fox, M., Lupi, M., Cabrera-Perez, I., Montanari, D. *In Review at Earth and Planetary Science Letters*. Tracing magma emplacement with surface and subsurface imaging.

## Recent Communications

**Invited talks**

China University of Geosciences (June 2026) - Using river network topography to understand the drivers of landscape evolution

Institute of Earth Environment, China Academy of Sciences (June 2026) – Modelling Biosphere-Geosphere Interactions with LandLab and LPJ-GUESS

TopoToolbox 3 Hackathon (March 2025) – Two-day meeting at University of Potsdam to contribute to development of TopoToolbox 3

British Society for Geomorphology Annual Meeting (September 2024) – Panel Discussion: Getting funded, navigating grant and fellowship applications

Georgia Institute of Technology (February 2024) – Exploring Earth's Dynamic Surface with Topographic Inverse Methods

**Oral presentations**

European Geosciences Union (May 2026) - Discerning the relative importance of Tectonic vs Climatic Controls on Topography from River Profile Analysis

British Society for Geomorphology Annual Meeting (September 2024) – Discovery of lithospheric drip explains topographic rejuvenation of the Uinta Mountains, USA (**Won best postgraduate talk**)

European Geosciences Union (April 2024) - Discovery of lithospheric drip explains topographic rejuvenation of the Uinta Mountains, USA

American Geophysical Union (December 2023) – Inversion of river networks reveals control of erodibility by bedrock-bedload contrasts

American Geophysical Union (December 2023) – Discovery of lithospheric drip explains active uplift in the Uinta Mountains, USA

**Science Communication**

The Evening Standard Tech and Science Daily Podcast - Two appearances as an expert in Earth Sciences

The World from PRX Radio Programme – Appearance as part of press release for coauthored work on Mt Everest (Xu et al. 2024)

## Teaching and Supervision

**Lectures and Tutorials**
Understanding the Climate Crisis (2026, UoG) – Development and delivery of 2 x 1 hour lectures, and 8 x 1 hour group exercises. Use of multimedia such as mentimeter and padlet to drive student engagement. 30 credit course delivered to ~200 first and second year students from social sciences, sciences, and humanities.

Physical Geography, Coastal Landscapes-Mass Movement (2025, University of Glasgow) – Delivery of 1 hour tutorials to groups of 6 or less first year geography students.

Tectonic geomorphology (2023, UCL) – Delivery of 1 x 2 hour lecture to 4$^{th}$ year Master's students

**Postgraduate Teaching Assistant**
Maps Images and Structures Scotland fieldtrip (2023, UCL)
Global Tectonics Oman fieldtrip (2023, Yale University)
Sedimentology and Stratigraphy Pyrenees fieldtrip (2023, UCL)
Tectonic Geomorphology (2022, UCL)
Sedimentology and Stratigraphy Wales fieldtrip (2022, UCL)
Matlab for Earth Sciences (2022, UCL)
Surface Processes (2021, UCL)

**Supervision and Mentorship**
*PhD Supervision*
Yonathan Leonel Viquez Carrera (2025-present, UoG), Xander Watson (2026-present, UoG)

*MSc Supervision Students*
Alex Walton-Keefe (2021, UCL), Chris Schverin (2022, UCL) and Anastasiia Kazakova (2023, UCL)

*BSc Students*
Andrew Lancaster (2025, UoG), Nasra Cadow (2025, UoG), Lily Kerby (2026, UoG)

*Mentorship*
Thesis writing mentor to Ella Boswell (University of Glasgow, 2026)
Research Experience Placement Mentor to Undergraduate Student Wasim Mustafa (UCL, July-August 2021)

## Technical Skills

**Laboratory Experience**
London Geochronology Centre
Physical rock crushing, heavy mineral separation (Zircon and Apatite), Apatite picking, single-crystal helium extraction, LA-ICP-MS
Natural History Museum
Thin section analysis, quartz cathodoluminescence spectroscopy, Scanning electron microscopy, fluid inclusion microthermometry, LA-ICP-MS
**Numerical Modelling**
Inverse and forward modelling of river networks, landscape evolution modelling (Landlab, FastScape), thermochronology modelling (age2edot), palaeotopographic reconstructions, isostatic adjustment modelling, dynamic vegetation modelling (LPJ-GUESS)
**Computing**
Microsoft Office (Word, Powerpoint, Excel etc), High Performance Computing, Matlab, Python, C++, GMT, QGIS, ESRI ArcGIS, LaTeX, Google Earth Engine, Inkscape, GIMP

## Professional Service

Served as a reviewer: PNAS, Nature Comms, Earth and Planetary Science Letters, Earth Surface Processes and Landforms, JGR: Earth Surface, Geophysical Research Letters, Global and Planetary Change, Tectonics, Tectonophysics, ESurf, Journal of Plant Ecology

UCL Earth Sciences PhD student departmental representative (2021-2022)

Member of the European Geosciences Union and British Society of Geomorphologists